

ABHAY RAJ RAGHUWANSHI

MCA Final Year

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SUMMARY

Aspiring Python and AI/ML engineer with a strong academic background in computer applications and hands-on experience in Python-based web application development. Skilled in database-driven systems, problem-solving, and collaborative project execution. Actively seeking opportunities to apply software engineering fundamentals in real-world applications.

TECHNICAL SKILLS

Programming & Libraries: Python, SQL, C, Numpy, Pandas
Machine Learning: Regression, Classification, Time-Series Analysis, Feature Engineering, Model Evaluation & Tuning
Databases: MySQL, PostgreSQL
Platforms and Technologies: Visual Studio, IntelliJ IDEA, Jupyter Notebook, Git, GitHub
Operating Systems: Windows

PROJECTS

Fitness Tracker – Time-Series Exercise Classification & Rep Counting

- Analyzed IMU sensor data (accelerometer and gyroscope) using signal processing techniques including filtering, FFT, and statistical feature engineering.
- Built machine learning models including Random Forest, SVM, and Neural Networks, achieving **80–95% accuracy** across five exercise types.
- Performed comprehensive EDA and created technical documentation with performance visualizations.
- Designed a custom rep-counting algorithm using peak detection and zero-crossing logic.

CortexFlow – Agentic GenAI Workflow Engine

- Designed a production-grade agentic AI system that decomposes complex tasks into executable workflows using an LLM-based planner–executor architecture.
- Implemented an end-to-end Retrieval-Augmented Generation (RAG) pipeline with document ingestion, semantic embeddings using SentenceTransformers, and FAISS-based vector storage with SQLite metadata persistence.
- Engineered a constrained LLM planner generating structured JSON plans with predefined tools and robust fallback mechanisms.

POSITIONS OF RESPONSIBILITY

Squad Leader — National Cadet Corps (NCC)

- Assisted in maintaining discipline, unity, and effective communication among cadets.
- Held responsibility for managing and coordinating squad activities during training camps.
- Demonstrated leadership, accountability, and organizational skills.

EDUCATION

Master of Computer Applications (MCA) SGSITS, Indore — RGPV University	Present CGPA: 8.09/10
Bachelor of Technology Oriental College of Technology, Bhopal — RGPV University	2024 CGPA: 8.35/10
Class XII (CBSE) Beersheba International School, Pipariya	2020 Percentage: 67.4%
Class X (CBSE) Beersheba International School, Pipariya	2018 Percentage: 70.8%

ADDITIONAL INFORMATION

Certifications & Achievements:

- NCC ‘C’ Certificate
- Solved 100+ coding problems across LeetCode, CodeChef and hackerRank.
- Python Certification – W3Schools
- International Level Taekwondo Player